MID Term Exam Solution

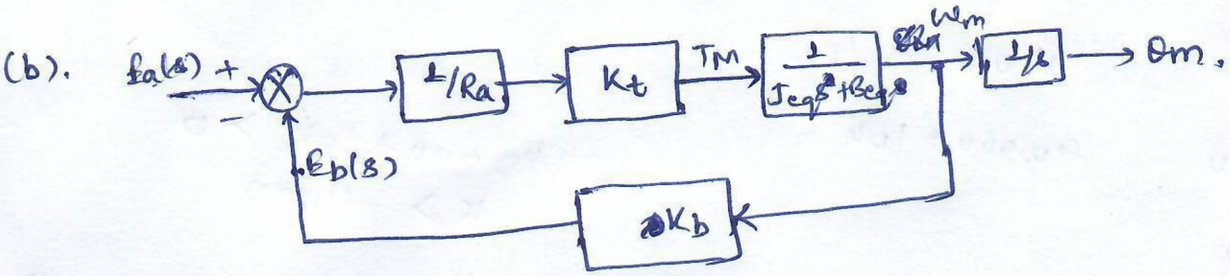
Q. 1

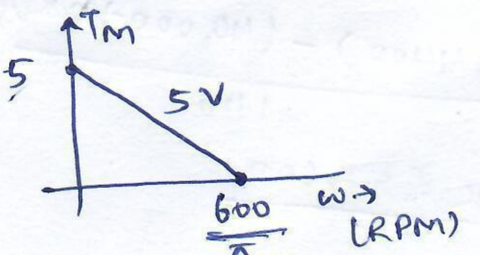
~~(a)~~. $J_{eq} = \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 J_4 + \left(\frac{1}{2}\right)^2 \times (2+2) + 1$

$= \left(\frac{1}{16} \times 32\right) + \left(\frac{1}{4} \times 4\right) + 1 = 2+1+1 = 4 \text{ kg-m}^2.$ 2

$D_{eq} = \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 \times 16 = \cancel{4}\,1 \text{ N-m-s/rad.}$ 2

(b).





$= \frac{\cancel{600}^{10}}{\pi} \times \frac{1}{\cancel{60}} \times 2\pi$
$= 20 \text{ rad/s.}$

$\because T_m = K_t i_a = K_t(E_a - E_b)/R_a$
$= K_t(E_a - K_b \omega_m)/R_a$
$T_M = \left(\frac{K_t}{R_a}\right) E_a - \left(\frac{K_t K_b}{R_a}\right)\omega_m.$
Tstall

so, $\frac{K_T}{R_a} = \frac{5}{5} = 1$ 2
5 → Rated vol.

& $\frac{K_t K_b}{R_a} = \frac{5-0}{20-0} = \frac{1}{4}, \Rightarrow K_b = 1/4.$ 2

so, $\frac{\theta_m(s)}{E_a(s)} = \frac{1}{s}\left[\frac{\frac{K_t}{R_a} \times \frac{1}{(J_{eq}s + B_{eq})}}{1 + \frac{K_t K_b}{R_a} \times \frac{1}{(J_{eq}s + B_{eq})}}\right]$

$= \frac{1}{s}\left[\frac{K_t}{R_a(J_{eq}s + B_{eq}) + K_t K_b}\right].$

$= \frac{1}{s}\left[\frac{K_t/R_a}{J_{eq}s + \left\{B_{eq} + \frac{K_t K_b}{R_a}\right\}}\right]$

$= \frac{1}{s}\left[\frac{1}{4s + \left(\cancel{4}\,1 + \frac{1}{4}\right)}\right] = \frac{1}{s\left[4s + \cancel{4}\,1.25\right]}$

$\frac{\theta_m(s)}{E_a(s)} = \frac{1}{s}\left[\frac{0.8}{\cancel{0.9411}\,3.2\,s + 1}\right].$ 4

(c). $\because \frac{\theta_L}{\theta_m} = \frac{1}{4} \Rightarrow \frac{\theta_L(s)}{E_a(s)} = \frac{0.2}{s}\left[\frac{1}{\cancel{0.9411}\,3.2\,s + 1}\right].$ 3

Q2.

~~At motor shaft,~~

~~$J_{eq} = 1 + \left(\frac{1}{2}\right)^2 \times 1 =$~~

At load side,

$$\boxed{\begin{aligned} J_{eq} &= 1 + (2)^2 \times 1 = 5 \quad Kg\text{-}m^2 \\ B_{eq} &= 0 + (2)^2 \times 1 = 4 \quad N\text{-}m\text{-}s/rad. \end{aligned}}$$

Say force at translational system be, $F(s)$, so,

~~&~~ Torque balance at load gear be,

$$\boxed{T_{eq} = \left(J_{eq}\ddot{\theta}_L + B_{eq}\dot{\theta}_L + K\theta_L\right) + \underbrace{F(s).r}_{\text{Torque from translational motion side}}}$$

also, $T_{eq} = \left(\frac{20}{10}\right) T_m = 2T_m$. & $\boxed{F(s) = [Ms^2 + s]X(s)}$

so, $2T_m(s) = (5s^2 + 4s + K)\theta_L(s) + \{(Ms^2+s)X(s)\} \times 2$

also, $\theta_L(s) = \frac{X(s)}{2}$

so, $4T_m(s) = \left[(5+4M)s^2 + \cancel{(4+2M)}\, 8s + K\right] X(s)$.

$$\therefore \boxed{\frac{X(s)}{T_m(s)} = \frac{4}{\{(5+4M)s^2 + 8s + K\}}}$$

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$$\frac{X(s)}{T_m(s)} = \frac{4/(5+4M)}{\left\{s^2 + \frac{8}{(5+4M)}s + \frac{K}{(5+4M)}\right\}}$$

$\because$ settling time $(t_s) = \frac{4}{\xi\omega_n} = 15$

$$\frac{4}{48/(5+4M)} = 15 \Rightarrow (5+4M) = 15$$

$$\Rightarrow \boxed{M = 10/4 = 2.5\ kg}$$

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Ques. 3:- $G(s) = \frac{K}{(s+100)} \times \frac{10}{(s+20)^2}$

$1+G(s)H(s)=0$; $1+\frac{10K}{(s+100)(s^2+40s+400)}=0$.

$s^3+40s^2+400s+100s^2+4000s+40,000+10K=0$.

$s^3+140s^2+4400s+40,000+10K=0$. — (1)

s^3	1	4400
s^2	140	$40,000+10K$
s^1	α	
s^0	$40,000+10K$	

$40,000+10K>0$
$K>-4000$

& $\alpha>0$

$\left\{\frac{(140\times4400)-(40,000+10K)}{140}\right\}>0$

$\Rightarrow K<57,600$.

So, range of 'K' should be,

$-4000<K<57,600$

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(b). Characteristic equation can be written as,

$(s+a)(s^2+2\xi\omega_n s+\omega_n^2)=0$.

assuming overshoot occurs ~~due~~ due to complex conjugate poles of $(s^2+2\xi\omega_n s+\omega_n^2)$, we have, $\xi=0.69$. 2

So, $(s+a)(s^2+1.38\omega_n s+\omega_n^2)=0$

$\Rightarrow s^3+1.38\omega_n s^2+as^2+(\omega_n^2+1.38a\omega_n)s+a\omega_n^2=0$. — (2)

Comparing (1) & (2), we have,

$(1.38\omega_n+a)=140$ — (3) and $(\omega_n^2+1.38a\omega_n)=4400$ — (4)

and $\omega_n^2 a=(40,000+10K)$ — (5)

by solving (3) & (4),

$\omega_n^2+1.38\omega_n(140-1.38\omega_n)=4400$

$\Rightarrow \omega_n=\underline{25.92}$ ✓ or $\underline{187}$ ✗

(this will make system unstable as 'a' will be -ve).

~~K=~~

~~so~~, $a = 140 - 1.38\omega_n$

$a = 104.23$ (Real pole), $\Rightarrow s_1 = -104.23$

and complex poles

$s_2, s_3 = -\xi\omega_n \pm j\omega_n\sqrt{1-\xi^2}$

$s_2, s_3 = -17.885 \pm j18.76.$ 4

$\boxed{K \approx 3002}$ - 3

also, $s_1 \approx 6 \times (\xi\omega_n)$, so, overshoot will be majorly due to complex poles & close to 5%.

Q4. $G(s) = \dfrac{2}{(10s+1)}$

at $t=0$, $e(t)=1$, so, $u(t) = 0.8$

$c(t) = 2 \times 0.8(1-e^{-t/10})$ $\quad (c_{final} = 1.6,\ c_{initial} = 0)$

a change in $u(t)$ will occur when, $c(t) > 1.2$, so,

$c(t_0) = 1.2 = 1.6(1-e^{-t/10})$

$\boxed{t_0 = 13.863 \text{ sec.}}$

Another switching of $u(t)$ from '0.8' to '0' will occur when,

$e(t) < -0.2$ or $c(t) < 0.8$,

so, $\quad 0.8 = 1.2\, e^{-t_1'/10}$ $\quad (c_{final} = 0,\ c_{initial} = 1.2)$

$t_1' = 4.054$ sec.

so; $\quad t_1 = 13.863 + 4.054 = \underline{17.917 \text{ sec}}$ ✓

$3 \times 3 = 9$

now, $u(t)$ will switch from '0' to '0.8' again when,

$e(t) > 0.2$, or $c(t) > 1.2$

so, $\quad c(t_2') = 1.2 = \underbrace{(1.6)}_{c_{final}} + (\underbrace{0.8}_{c_{initial}} - \underbrace{1.6}_{c_{final}})(e^{-t_2'/10})$

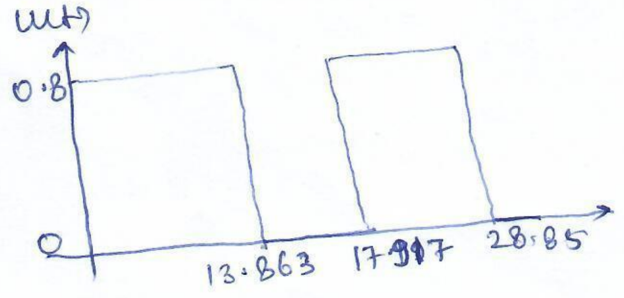
$t_2' = 6.931$ sec.

so, $\quad t_2 = 17.917 + 6.931 \approx \underline{28.85}$ sec.

(b) Period of oscillation $= (4.054 + 6.931)$

$= 10.985$ sec. $\quad -3$

(c).



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